

eleID and AsianElephant.info

An Automated Elephant Recognition System for collaborative Habitat Monitoring

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Corea²

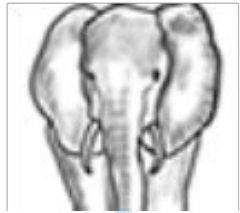
¹Department of Electronic and Telecommunication Engineering,
University of Moratuwa, Sri Lanka

²Sri Lanka Wildlife Conservation Society, Dehiwala, Sri Lanka

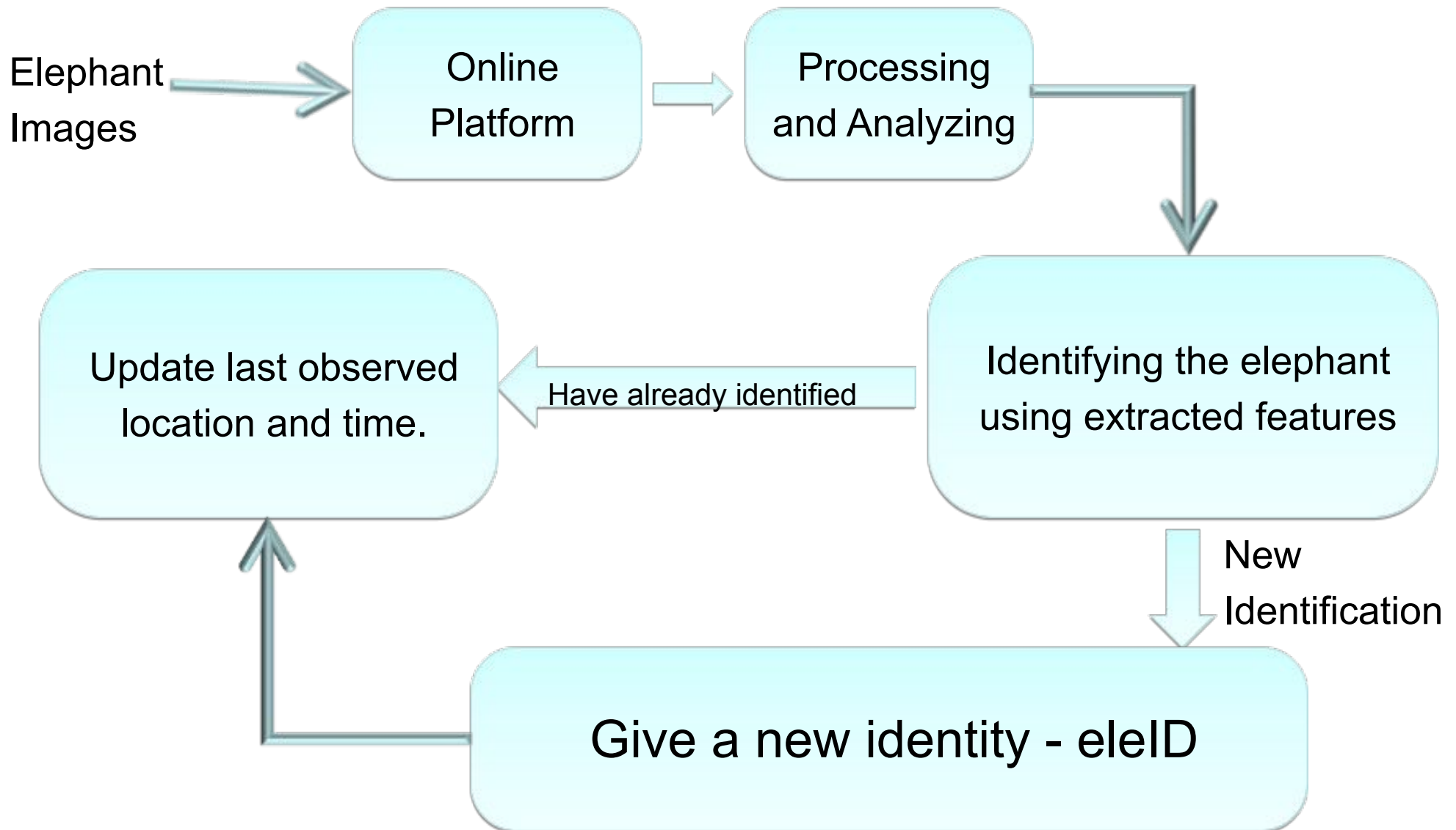
OBJECTIVES...

- **Efficient mechanism of habitat monitoring and distribution calculation**
- **Faster, less expensive and non-invasive mechanism**
- **Promote Citizen Science**
- **Google for Asian elephant information**

Automated Elephant Recognition System



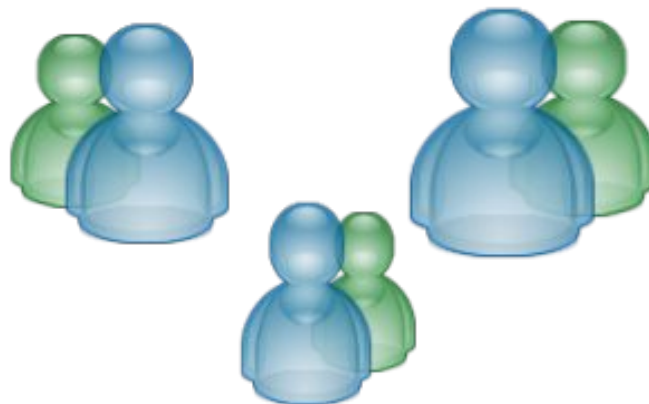
Methodology



elePortal



- Interface for the users all around the world.
- Easy access to individual elephant information.
- Promote Citizen Science.



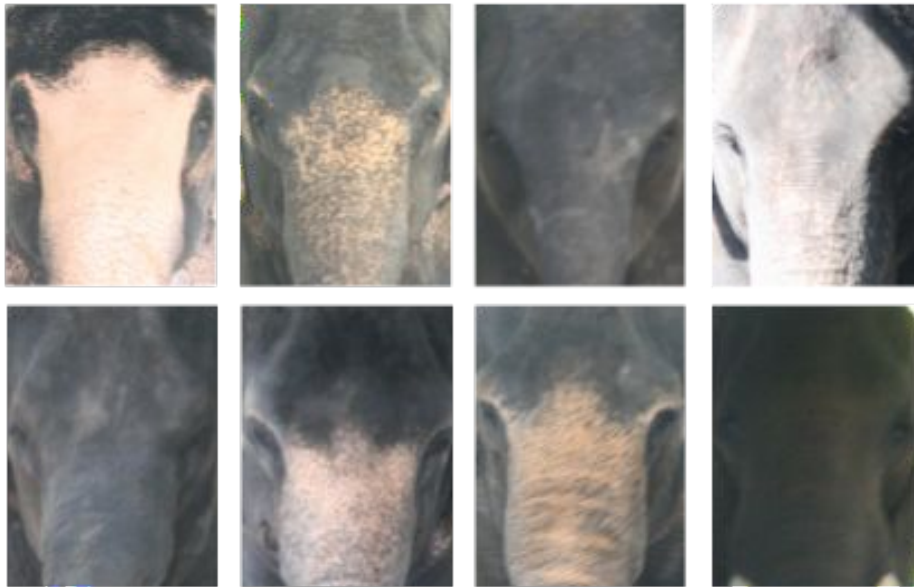


- Automatically identify individual elephants from given images and data
- Initial implementation based on [Principal Component Analysis](#)
- Need frontal images
- Further developments are in progress to make the system robust

eleID *cont...*



- Prior to the identification already identified elephant images should be given to the system

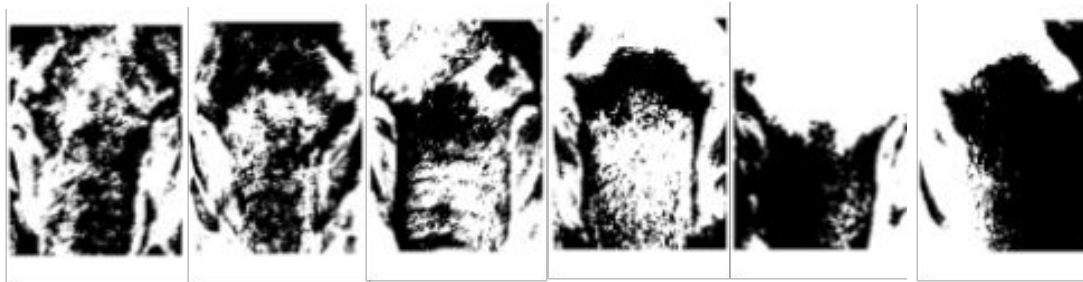


- Along with images other information are kept in the database

eleID *cont...*

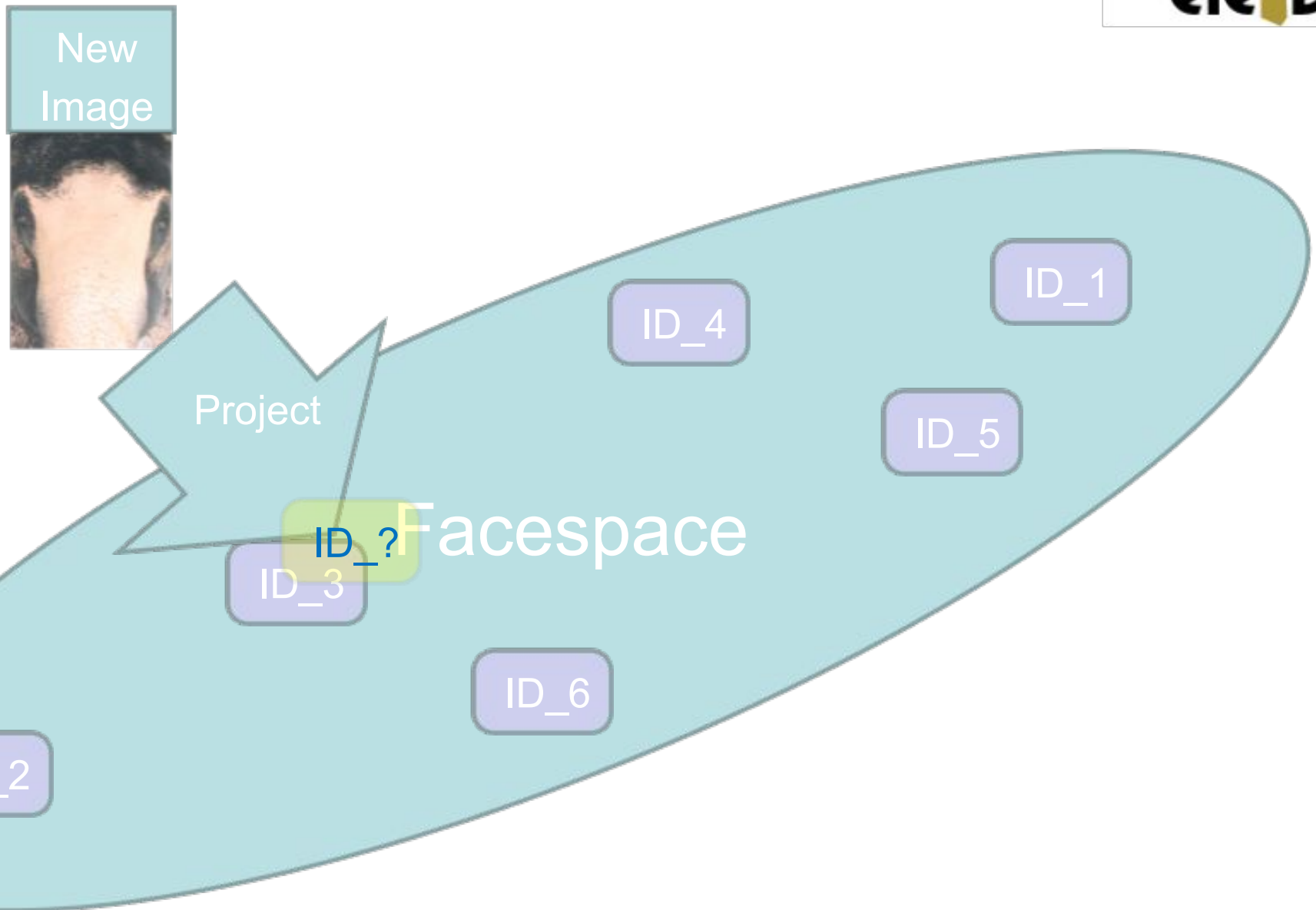


- Eigenfaces of elephant face images



- New images given for identification are projected on to the facespace

A simple model...



eleID *cont...*



- Best match for the input image will be given.



- Other matches are used to verify the result

Brick-walls



- When pose of the elephant face changes, it affects the accuracy of identification
- Have to utilize other information effectively in the identification mechanism
- When illumination varies a lot, it affects the accuracy.

Current work



- To handle pose variations in elephant face images, presently AAM - Active Appearance Models are being tested .

Future direction



- Utilize additional information actively in the identification mechanism
 - Evidence Combining Mechanism.

Eq. Far data



Summary

- Individual elephant identification is important for habitat monitoring.
- Computer Vision based mechanisms can be successfully utilized for individual elephant identification.
- elePortal enables large community involvement and efficient data handling.

Acknowledgement

- Partially-funded by Tides Foundation through Google Geo Challenge Grant.
- Sub-project of Sri Lanka Wildlife Conservation Society's Saving Elephants by Helping People Project.

Q & A

A living geo-portal for Asian Elephants

- Starting point -> www.AsianElephant.info
- Google Earth Maps will be available in the home page
- Users can interact with them

Asian Elephant



SEARCH

Search this site:

SEARCH

HOME

ABOUT US

WHO'S NEW



[abc1](#)



[Administrator](#)

WHO'S ONLINE

There are currently 1 user and 0 guests online.

Online users



[Administrator](#)

ACTIVE FORUM TOPICS



[Global warming](#)

[more](#)

ADMINISTRATOR

- [Build a GMap macro](#)
- [My account](#)
- [Node locations](#)
- [User List](#)
- [User locations](#)
- [Activity](#)
- [Create content](#)
- [Administer](#)
- [Log out](#)

Gmap macro creation



The site will start with a map of Asia and zoom into the area of interest

- » Activity
- » Create content
- » Administer
- » Log out

POLL

What do you think about the elephant count in Sri Lanka. Is it increasing?

Yes



67%

No



33%

I don't know really



0%

Total votes: 3

[Add new comment](#) [Older polls](#)

NEW FORUM TOPICS

- » Global warming

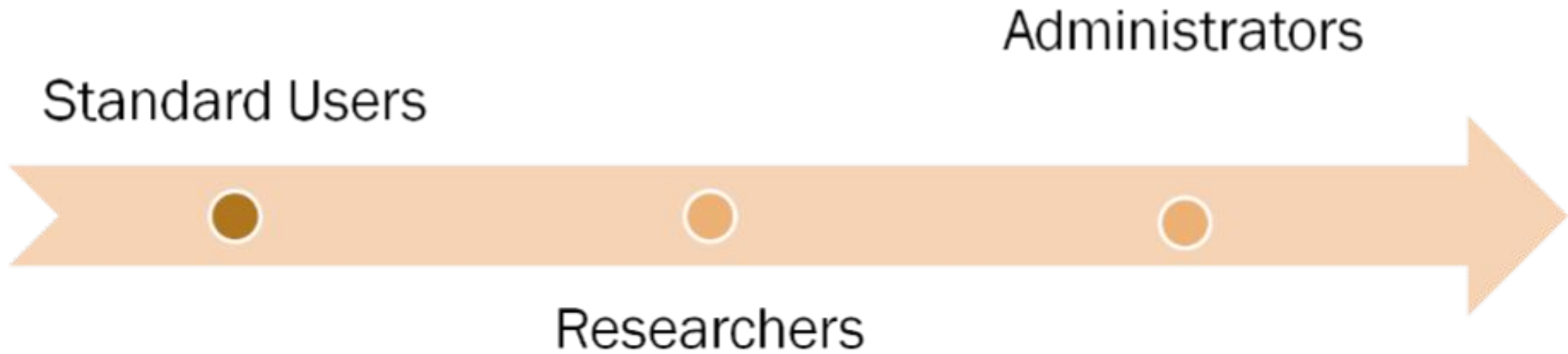
[more](#)

Gmap macro creation

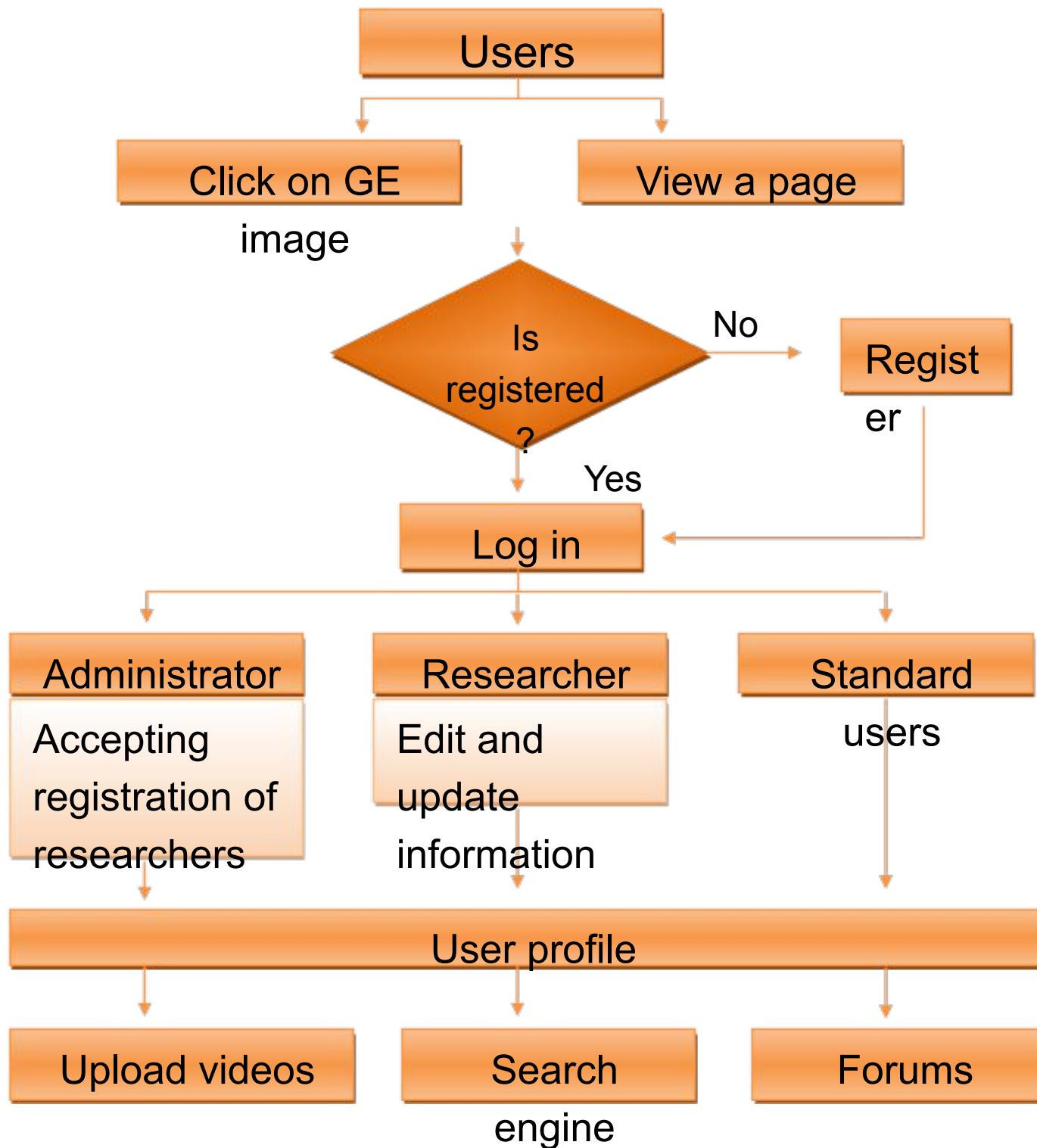


Users can zoom to the level needed

Types of users



Different types of users have different accessibility levels.



- First, all the users should log in with unique username/password
- Without logging in to the system they are not allowed to see the relevant data regarding researches, forums, etc.

- The first time users can register for free.

Standard users are being registered by an automated system, while for **Researchers**, access is granted only after a review of credentials by project staff (**Administrators**).

Standard users

CAN

- create their own profiles and update them.
- upload videos and images onto the Geo Portal.
- share their experiences visually with the forum.
- search the database for information.
- create a shopping cart to add relevant information that can be accessed at any time

Researchers

CAN

- perform all the events which the 'Regular users' can perform
- draw a polygon covering project areas and then enter information about the data they have for the area and contact information.
- directly import Meta data from shape files imported through a semi-automated process.
- modify or update existing data.
- check stats of who has requested their data.

ADMINISTRATOR

- Build a GMap macro
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- Activity
- Create content
 - Blog entry
 - Forum topic
 - Group
 - Information form
 - Newsletter issue
- Page
- Poll
- Story
- Administer
- Log out

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What do you think about the elephant count in Sri Lanka. Is it increasing?

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67%

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33%

I don't know really:

0%

Total votes: 3

[Add new comment](#) [Order](#)

SECONDARY LINKS

- Researches

Create Information form

Title: *

Menu settings

Family / Last Name: *

Other Names: *

Title:

Position:

Organization / Institute:

Switch to rich text editor

The ID for excluding or including this element is: edit-field-organization-0-value - the path is: node/add/informationform

Telephone:

Email:

Fax:

Information of users can be collected through forms and searched by others. Users will also be able to update their profiles, add CV's etc and reply to Job postings

Search

Users can search the database using any of the following keys through a GE-API available in the website.

- By entering coordinates (WGS84) or a specific location name
- By manipulating GE and clicking on an area
- By using a drop down list up to divisional level in range areas
- By uploading a .kml or .shp file in WGS84 projection
- By typing search keywords to generate a map if the requested information exists

For example if we take the above map of Sri Lanka ;

- When a user draws a polygon on the map
 - It will list the research information of that area.
(Date / Organization / Project Name)

ADMINISTRATOR

- Build a GPS macro
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SECONDARY LINKS

- Researches

POLL

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Add new comment Older polls

NEW FORUM TOPICS



Users can create polygons around areas they are interested in and Researchers can do the same and enter data/published info about their work in the area.

Polls can be conducted on important current issues

- Also.....

When someone searches for people working on electric fencing

- The map will come up to show all locations where this work is reported.
- People can click through to get details and download project reports etc

Search this site:

SEARCH

ABOUT UPL

 abc1

✓ abc1

 Administrator

There are currently 1 user and 0 guests online.

Online users

 Administrator

✓ Global warming

- Build a GMaps macro
- My account
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- Administer
- Log out

- Researches

- Department of Wildlife Conservation, 2002 Survey of Elephant Density. Contact Director Research, www.DWLC.lk.
- Sri Lanka Wildlife Conservation Society, 2004 Survey of HEC. Contact www.SLWCS.org or info@slwcs.org
- Elephant Radio Collaring Study - Center for Conservation Research, Contact _____

A search then identifies the locations and existing data sources for the area of interest along with contact info and researchers can edit their own information.

[HOME](#) [ABOUT US](#)

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SECONDARY LINKS

- » Researches

Environmental issues

Post new Forum topic:

Topic	Replies	Created	Last reply
 Global warming	0	5 weeks 17 hours ago by Administrator	n/a

ASIAN ELEPHANT NEWSLETTER

Stay informed on our latest news!

User: Administrator

[UNSUBSCRIBE](#)

Previous issues

- 
- 

Clients/Users can post and discuss anything about Asian Elephants and get answers by Experts volunteering with AsianElephant.info

Clients/Users can subscribe to a newsletter

- Elephant identification software will also be interfaced with Google Earth through the Geo portal.
- This will be a “one stop shop” dynamic Geo Portal for all information pertaining to Asian elephant conservation and rese



Back Up

Unique Features

- Tail features.
 - Length of the tail
 - Nature of the tail tuft
- Ears.
 - Primary and secondary folds.
 - Ear bottom appearance using contour of an elephant ear.
 - Rips, holes and notches.
- Facial features.

Mathematics behind Eigenfaces

- Arbitrary image can be compressed using;

The diagram illustrates the mathematical formula for compressing an arbitrary image using Eigenfaces. The formula is presented as $\tilde{\mathbf{x}} = \mathbf{m} + \sum_{i=1}^M a_i \mathbf{u}_i$. Each term in the formula is linked to a descriptive label via a light blue arrow. The label 'Compressed Image' points to $\tilde{\mathbf{x}}$, 'Mean Image' points to $\mathbf{m}$, 'Eigenfaces' points to $\mathbf{u}_i$, and 'Coefficients unique to the image' points to a_i . The labels are contained within colored rounded rectangles: light blue for 'Compressed Image', light purple for 'Mean Image', pink for 'Eigenfaces', and yellow for 'Coefficients unique to the image'.

Compressed Image

$$\tilde{\mathbf{x}} = \mathbf{m} + \sum_{i=1}^M a_i \mathbf{u}_i$$

Mean Image

Eigenfaces

Coefficients unique to the image

Finding Eigenfaces

- Have to analyze the scatter matrix **C**,

The diagram shows the formula for the scatter matrix C with three annotations:

- A light blue box labeled "Number of training examples" has an arrow pointing to the N in the denominator.
- A light green box labeled "Identified Images" has two green arrows pointing to the x_j terms in the vector $(x_j - m)$.
- A light purple box labeled "Mean Image" has two light blue arrows pointing to the m terms in the vector $(x_j - m)$.

$$C = \frac{1}{N} \sum_{j=1}^N (x_j - m)(x_j - m)^T$$

Cont...

Finding Eigenfaces *cont...*

- Apply Eigenvalue decomposition;

$$\mathbf{C} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T = \sum_{i=1}^N \lambda_i \mathbf{u}_i \mathbf{u}_i^T$$

The diagram illustrates the Eigenvalue decomposition equation $\mathbf{C} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T = \sum_{i=1}^N \lambda_i \mathbf{u}_i \mathbf{u}_i^T$. Three callout boxes with arrows identify the components: a yellow box labeled 'Scatter matrix' points to $\mathbf{C}$; a light purple box labeled 'Eigenvalues' points to λ_i ; and a light green box labeled 'Eigenvectors called as *'Eigenfaces'*' points to $\mathbf{u}_i$.

Scatter matrix

Eigenvalues

Eigenvectors called as *'Eigenfaces'*

Identification

- Finding the coefficients unique to the frontal image of the elephant;

